Perfect Lift Gravity Counterbalance Explanation (Continuous)

Goal: To negate the force of gravity by using springs to offset the weight of the lift. At any point of extension on the lift, the lift should fully counterbalance itself. Additionally it should significantly reduce current draw required to power the lift, (that can go to powering other subsystems quicker).

Use Cases: In my opinion good fundamental design is about maximizing simplicity and only adding complexity when/where you really need it. Remember most of the time lift counter springing is probably unnecessary complexity and should be a low priority, though it is highly dependent on the game.

- Usually it's going to boil down to - do you have a speed and/or current draw bottleneck?
  - Heavy load you have to lift up but you don't want to compensate with slow slides and high current draw (Like 12791 and 8644 in Powerplay)
    - 12791 and 8644 were both “Crane” style robots that had their horizontal extensions mounted on their vertical extension, with 8644 having a motor and 5 servos among other heavy components.
  - Extremely fast slides required and you cannot afford to spend any significant amount of your current (amps) budget on your slides.

Understanding Continuous Stringing Forces: The force or torque that the motor must exert to lift up the extension changes as each stage of the slide fully extends and becomes static.

On typical continuous lifts, stages extend one after another, from inside to outside. This means that as time goes on and the lift extends more, the mass being moved increases and so does the amount of torque required from the motor.

There are two key forces at play: Inertial Torque (Acceleration) and Gravitational Torque (Lifting Weight). Simply put:

- Inertial Torque = the torque needed to accelerate the mass upwards
- Gravitational Torque = the torque of gravity pulling the lift down
  - Remember: If you lift is at an angle / not parallel to gravity then the effect of gravity is less, meaning you would need weaker springs (Multiply by sin theta).

Trends:

- Inertial Torque: As the lift's stages extend (assuming the inside slides to the outside is the order and full speed is being applied) the inertial torque increases until the lift hits its end and stops, the inertial torque then dropping to zero when at a standstill.
- Gravitational Torque : As the lift's stages extend (once again assuming that inside to outside is the order) then more torque is required for each stage fully extended until the total gravitational torque is reached (Total Fgravity = total mass of your lift x 9.81 m/s^2)

Side Tangent - Introduction To Counterspringing Inertial Torque:

By perfectly counterspringing your lift you would eliminate the gravitational torque. There is no easy way to counterspring inertial torque, (it would be like trying to counterspring a horizontal extension).

- If you wanted to counterspring inertial force (which only exists during acceleration) you would need a spring that would activate only during the acceleration and change forces during the acceleration.
- There are two main categories for doing this:
    - Passively: Normal Hooke-ian springs cannot model the acceleration force profile, meaning you would need a specialized nonlinear / variable-rate spring or some sort of cam / linkage system that transforms a springs force.
    - Actively: Using a servo or some sort of actuator to pull a spring to match the acceleration force profile.

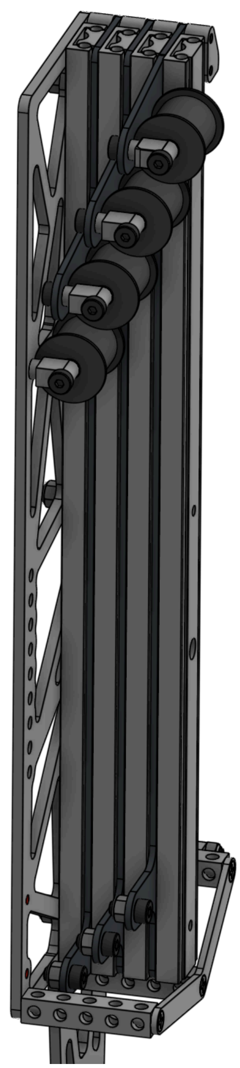
Note: I only mentioned this because I found it interesting, not because I think it is a good idea to spend your time doing it. To my knowledge nobody has ever actually tried to do this in any robotics competition, since the benefits are marginal at best, especially for the effort.

Understanding Constant Force Springs: These are pieces of stainless steel strips (think 0.05-0.15mm) that have been wrapped into a coil that unwraps to deliver force. They are formed by “pre-stressing” the steel or putting the steel under tension then wrapping it onto itself or a drum. Because the material is so thin the radius of each circle is nearly the same, making the force ***nearly*** constant. I say nearly but the difference is negligible (usually only ~1-2% difference in the force exerted by different wraps of the spring).

Understanding Why Different Springs On Each Stage:

- Even though total gravitational torque does not change, the lift builds its way up to the total gravitational torque as it continues to extend stages. When a stage of the slides is completed and a new one is started the gravitational torque increases, since the mass that is being moved is increasing / the mass that can be under the effects of gravity is increasing.
- Therefore as the lift gets heavier (adding moving mass) the springing must compensate for the increases in gravitational torque.

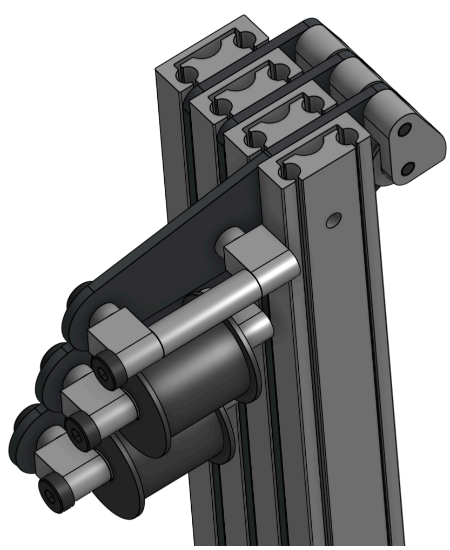
Now let's look at an example.



Example Explanation: The heaviest spring starts on the outside and is fixed to the side plate (pitch plate in this example but for robots with static slides you would need it to either attach via an outside insert or the inside drivetrain plate). The first / heaviest spring sits on a 3DP spool then attaches to the first sliding stage of the Misumi via the hole that comes in all of the constant force springs. This continues all the way to the last stage where the spring gets screwed into the plate that attaches to the Misumi for the outtake.

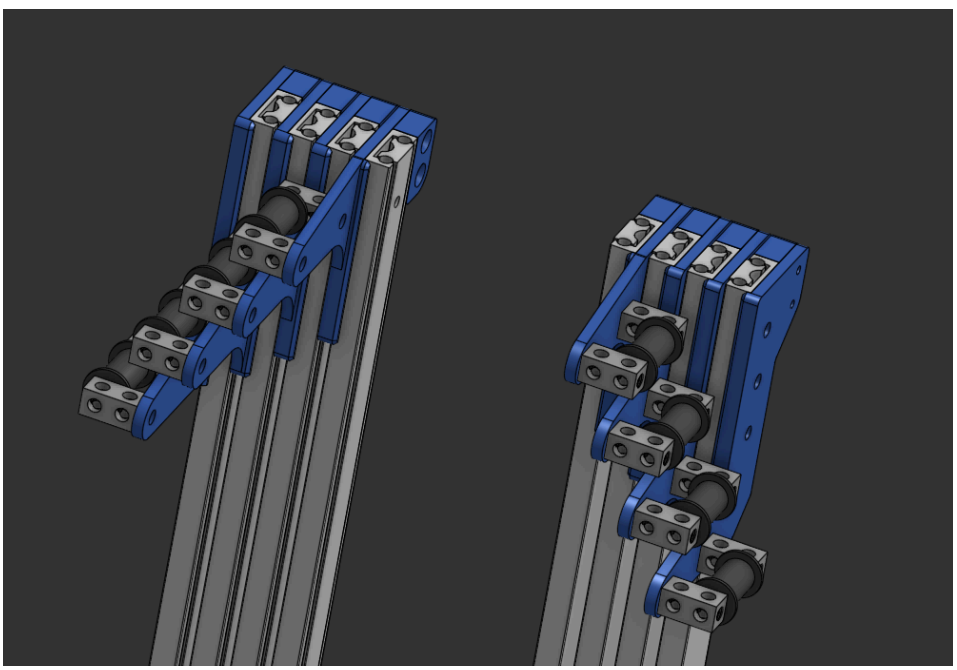
Example Execution and Reasoning: The springs are staggered so the spools do not interfere. In the image below we use GoBilda single block mounts and a 20mm spacer to create a super rigid and fully supported shaft for the spool to sit on.

- One thing to note is that double threading a screw / screwing a screw through two sets of threads is typically looked down upon, but it worked for us in this case. If you are worried that this might cause the screw to get stuck you can just drill out the first single block mounts threads.
- We used full metal spacers since we have had issues with 3DP spacers getting crushed when tightened, so we use metal spacers where we can since it makes it more rigid for a negligible amount of weight gained.
- As for the 3DP spool, they have flanges just to make sure the spring does not for some reason wrap in a weird way though we have not tested without them, so they may be unnecessary if the spring is getting extended without any twisting. In order to make assembly quick we printed the spools in two parts so that we would not need to unravel the spring just to put it onto the spool. Instead we just slid it on making assembly super easy and quick.
- Bearings can be unnecessary weight and complexity so we just went with a slightly oversized hole, which creates negligible friction, on par with what a bearing would generate, though a bearing can still be a viable option.
- There are countersprings on both sides, so that the force of counterspringing is equally distributed, though it could work on a single side too if you're confident in the rigidity of how you have mechanically linked the slides.
    - Additionally there are a limited number of springs out there with only specific weights. Typically there are more lower weight springs making it generally more accurate to use lower weight springs.



Design Flaw & Alternative Design: The current setup with the spools on the inside of the plates has an interference with any plates mounted to the Misumis on the spring's side. By switching the side of the plate that the spools are mounted to, this fixes the issue.

- Additionally this design uses beams with a standoff instead of the block mounts and a spacer. You would need to countersink one of the beams though.



Video: CountersprungLiftIRL.mov

- **What I'm doing**: I am moving the lift to random points that I want and letting it counterbalance. Though my hand is close and might look like it's stabilizing it at times, it's not. As soon as I move the lift to the position I release the lift and stop touching it.

- **Note**: There is not a spring on the last stage because the team never got to it before the season ended, no other reason.

Calculating Springs Needed: Calculating happens in two ways. With an approximation step, then a confirmation step. Though I am fairly confident you could do both or pick either and it would still work alright.

- **Approximating**:
  - First using Onshape, as long as you are using the FTC Parts Library, all COTS parts should have accurate part weights, same goes with using Onshape material library for plates (you can get reasonably accurate carbon fiber stats off the CNCmadness, SCS or wherever else you order). For 3DP I use Julia's material library and the "3DP PLA" option, which seems to be fairly accurate for me, but usually the 3DP parts are going to be the least accurate and if you really care you can use a slicer's weigh approximation or weigh them IRL (since most slicers can be wrong too).
  - You should find the weights of your:
      - Outtake (Arm, claw and whatever plates your using to mount to the Misumis of course, screws if you want to be as precise as possible)
      - Inserts (Alu / CF / 3DP inserts, you can also include the weight of the spring in final calculations)
  - Based on the weight of you outtake + # of Misumis and inserts you have, add it up to calculate the total weight of your outtake. That will be the first spring weight you need (divided by two if you are counterspringing on both sides). Subtract the weight of 1 set of Misumis and slide inserts to get the next spring weight, etc; until you reach the number of springs you need.
  - **I recommend approximating roughly at least** to make sure that your spool sizes are at least far apart enough to work. You don't want to underestimate and then realize that your spools are too large and too close to each other and you have to fix and remanufacture the inserts.
- **Calculating Exactly**:
  - The most accurate way to calculate the springs you need is when you do it IRL once the lift is fully assembled on the robot, strung and wired up. Ideally the lift is not connected by any form of transmission to the motor (since the motor and transmission has a bit of friction or tension). You take a force sensor or a luggage scale and extend the slides to measure the force on each stage.
  - Attach the sensor to the lift (usually it has some sort of hook and you can just attach that to a shaft or something). Start lifting the first stage at any point where the Misumi has not reached its endstop. Hold the first stage down now - someone can just put their hand over it to make sure it doesn't extend, and extend the second stage up - you are just weighing everything minus the first stage now / that is the goal. Continue until you have weighed all of the stages.
  - We measure 2-3 times and have different people do the measuring just to ensure we are getting the correct data. You can also compare to the expected value as

well as the change between the weight of the measurements should be basically the same as the CAD. Our measurements were only off about ~+-0.1-0.2lbs.
    - Remember it's always better to underestimate what springs you need.

Buying Springs: We tried to search for our springs on Amazon, and while they have the springs they seem to be a bit pricier than what McMaster Carr was offering, though this might depend on the specific spring you are searching for. Overall, McMaster Carr was the quickest option that was the most trusted and reliable, though there are other spring-specific vendors out there too that might work.

- McMaster Catalog: Constant Force Springs

Video of Lift When Powered: CountersprungLiftPowered.mov